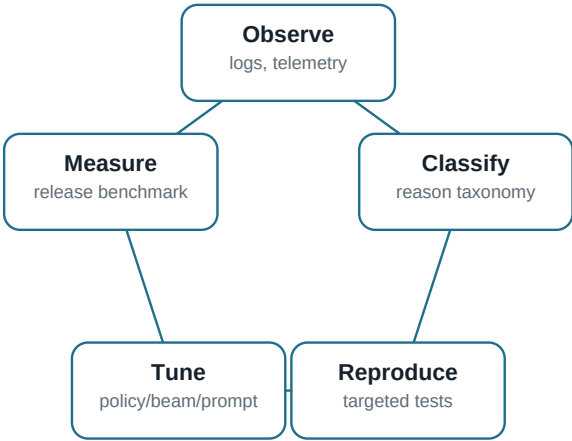


Observability, testing, and maintenance

A tab-autocomplete app needs maintenance loops: why did it show this, why did it suppress that, and where did latency go? KeyType has a more complete closed loop for those questions. Cotabby has good pure-function tests and debug tooling, but its durable observability is thinner unless launched with the debug flag.

Maintenance dimension	KeyType	Cotabby
Module boundaries	Local SwiftPM packages: AutocompleteCore, Prompting, ModelRuntime, ConstrainedGeneration, TokenProfiles, CompletionUI, TextInsertion, AppCompatibility.	Single app target organized by App, UI, Services, Models, Support. External CotabbyInference package from branch main.
Quality logs	predictions.log records PREDICT, SHOWN, SUPPRESS(reason), ACCEPT(word/full); telemetry counts generated/shown/suppressed/accepted and latency.	SuggestionDebugLogger emits prompt/raw/normalized blocks only when -cotabby-debug is enabled; latest latency and previews live in coordinator state.
Latency tests	Release package tests profile production path, cold prefill curves, branch-width reuse, and model runtime behavior.	Cache-hint tests and runtime logging exist; no equivalent retained aggregate benchmark in the main repo found.
Prompt tests	Prompting tests cover section ordering, budgets, FIM-related prompt behavior, etc.	PromptPolicyTests and LlamaPromptRendererTests cover prompt structure, cache hints, FM prompt drift harness gated locally.
Decision trail	Docs/05 ADR log plus playbooks for quality, performance, app compatibility.	ARCHITECTURE and AGENTS provide good maintainer guidance; fewer formal ADR-style decision records.
Dependency risk	Vendored llama.xcframework inside ModelRuntime; app catalog points to downloadable GGUF and generated ACPF profiles.	Xcode project references cotabbyinference via branch main, with local checkout 6b26200. Branch dependency can move unless resolved externally.

Maintenance loop maturity



KeyType has all five loop components in durable form.

Cotabby has Observe mainly in debug mode and Measure mainly via ad hoc tests or logs.